

Design and Analysis of Steel Support Structures Used in Photovoltaic (PV) Solar Panels (SPs): A Case Study in Turkey

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Abstract

As one of the most common and imperative contributing factors to clean energy aspect, solar energy takes a significant role around the whole world. Among neighboring countries regarding the energy sources, Turkey has a relatively more potential for solar energy to decrease its energy dependence to the other countries and to increase awareness for sustainable, easily reachable, economical and continuous energy use. In the photovoltaic (PV) solar power plant projects, PV solar panel (SP) support structure is one of the main elements and limited numerical studies exist on PVSP ground mounting steel frames to be a research gap that has not been addressed adequately in the literature. In this paper, aiming to provide a contribution to this gap, a PVSP steel support structure and its key design parameters, calculation method, and finite element analysis (FEA) detailed with a case study on a solar power plant in Turkey are described to obtain actual demand of environmental effect like loads wind, snow, and seismic loads conforming with Turkish codes and standards. FEA is done by using load calculation with creating model in SAP2000 and followed by analysis to determine maximum von Mises stress distribution on the PVSP steel support structure.

Keywords: Photovoltaic (PV), Solar Panel (SP), Steel, Support Structure, Structural Design, Finite Element Analysis (FEA)

1. Introduction

Solar energy is a hopeful, sustainable, new kind green energy which is never-ending, independent and plentiful. Solar panels (SPs) can be various cross-sections (e.g., square, rectangle) and sizes but their main purpose is to convert the sun light in order to make electricity. Normally, solar power systems can be separated into three used groups like (i) concentrating solar power, (ii) solar-thermal absorbers and (iii) photovoltaic (PV) SPs. PVSPs directly transform solar to electrical energy using semiconductor materials which can produce free electrons utilizing of sunlight energy (Parida *et al.*, 2011). PVSPs have many usage fields, such as solar home systems, PV water pumping, remote building, solar cars and airplanes, satellites and space vehicles, etc (Kalogirou, 2004). Such a wide range of application fields increase demand for PVSPs due to the electrical power generation through PV transformation gives clean, safe and efficient way of supplying energy. The developments of current technology for PVSP field significantly vary, especially for installation methods and mounting locations (e.g., ground, roof, or integrated with the building envelope) affected by wind, snow and seismic loads differently. Experimental and numerical studies have been conducted by several researchers to show the effectiveness of different types SPs. Mihailidis *et al.* (2009) proposed a finite element analysis (FEA) of two different design approaches of SP support structures such as fixed support and adjustable support structure design. Cao *et al.* (2013) performed a wind tunnel experiment to evaluate wind loads on SPs mounted on the flat roofs. Lin *et al.* (2013) proposed a FEA approach to find the structural deformation and misalignment of solar radiation using the effects of self-weight and wind loads. The result shows that this technique was found to be sufficiently reliable to design PV systems. Aly and Bitsuamlak (2013) carried out experimental tests on large civil engineering structure models with geometric scale of 1:500 to 1:100 to produce an aerodynamic model of SP subjected to wind load and mounted on ground. Mathew *et al.* (2013) studied on design and stability analysis of SP support structure made of mild steel. The result shows that the SP support structure can able to sustain a wind load with velocity 55ms^{-1} . Baetu *et al.* (2013) presented numerical simulations performed in ANSYS for analysis of wind action on SPs located on a flat roof with and without parapets to understand the location of mounting of SP support structure affects the SPs' performance. In recent years, the design of PVSP support system has become the focus of attention with the popularization and increasing application fields.

The PVSPs are typically installed on aluminum or galvanized/ painted/ stainless steel support structures (the ground mounting steel frame). The construction of solar energy systems, mainly steel materials have a favorable custom in structural engineering applications, but the aluminum alloy is increasingly being used due to its characteristics such as lightweight, high strength-to-weight ratio, good corrosion resistance, durability, recyclability properties, high strength similar to steel and excellent formability (e.g., extrusion) (Girard *et al.*, 2016; Drury *et al.*, 2014; Hernandez *et al.*, 2015; Avci-Karatas and Celik, 2017; Avci-Karatas *et.al.*, 2018). Wang *et al.* (2018) studied on the actual project case design and optimization of fixed PV support structure using Japanese Industrial Standard based on SAP2000.

It has been observed that PVSP is one of the fastest growing industries in the world. Turkey sits in a favorable status in the middle east and southeast Europe regarding solar energy potential (Dawood, 2016). 7.5TWh was generated in 2018 using solar power which was 2.5% of Turkey's electricity (Ministry of Energy and Natural Resources of Turkey, 2019). Installed PVSP capacity was 5GW, with the energy ministry planning to have another 10GW installed in the 2020s (OECD, 2019). Since the usage of solar energy are more attractive to investors and have recently become the focus of considerable interest, the design of PVSP support structures has merit in structural engineering applications. So far, to knowledge of the author (Avci-Karatas, 2019a; Avci-Karatas *et al.*, 2019b; Avci-Karatas and Celik, 2019c) limited guidance exists to help the engineers to design of PVSP steel support structure using Turkish codes and standards in the current technical literature. This consideration motivated the research presented in this paper to search, design and analysis of PVSP steel support structure under the wind, snow, and seismic loads specified according to Turkish codes and standards to make a contribution to a gap in a relatively recent development in the field of PVSP technology for developers and investors. For this purpose, an example on a PV solar power plant project in Turkey was considered to provide quotative data to describe the results for the currently designed, modeled and analyzed of the PVSP support structures. SAP2000 v14 (2009) software was used in this paper to carry out the design, FEA and research on the bearing capacity of the PV support structure under various load conditions using Turkish codes and standards.

2. Description of PVSP Steel Support

When it comes to structural design of support structure for SPs, many different factors should be clearly considered in the design stage of support structure sitting on the ground. The local and site-specific investigation of climate data are important to properly decide on structural design parameters for PVSP steel support structures (Celik and Celik, 2019).

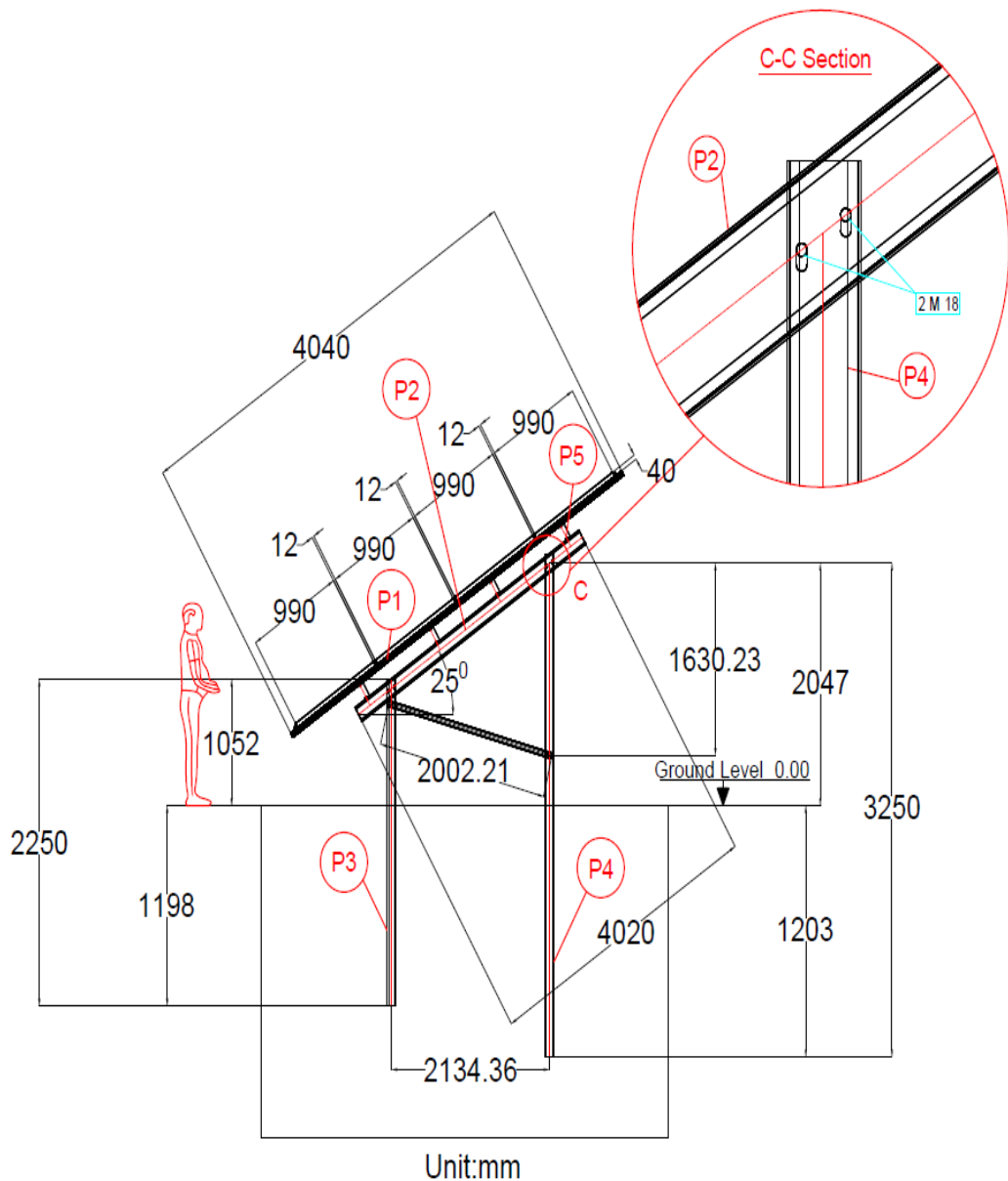
The power plant project with 500kWp total kilowatt-peak capacity is in Siirt province of southeastern Turkey. The system will generate 683,427kWh of solar energy in one year. The mounting angle of PVSPs support is $\alpha = 25^\circ$. Totally, 44 PVSPs having the size of 1650mmx990mmx40mm are used as 4 rows and 11 columns.

3. Materials

With the 4 rows and 11 columns PVSPs, the ground mounting steel frame has five basic bearing members named as “rail” for PVSP mounting, “beam”, column”, “purlin”, and “brace”, respectively. Figure 1 shows the general views of PVSP steel support structure. The basic design parameters and material properties are summarized in Table 1 and Table 2, respectively. The profiles are designated as P1, P2, P3, P4, P5 and P6 for rail, beam, front column, back column, purlin and brace, respectively (Figure 1 and Table 2). The total length and width of PVSP frame are denoted as (L) and (W_F) in the Table 1.

Table 1. The design parameters of PVSP ground mounting steel frame

Parameters	Values
PVSP dimensions (mm)	1650x990x40
PVSP weight (kg)	18.5
Mounting angle of PVSP α (degree)	25^0
PVSP average height from the ground (mm)	~1500
Row number of PVSP	4
Column number of PVSP	11
Number and length of rail profiles (mm)	22x4040
Number and total length of beam profiles (mm)	5x4020
Width of frame, W_F	2134.36
Number of supported span and total length of frame, L (mm)	4x4537=18148
Total covered area of PVSP, A_{PVSP} (m ²)	4.040x18.148=73.32
Total height of front/short column (mm)	2250
Total height of back/long column (mm)	3250
Height of front column profiles above ground level (mm)	1052
Height of front column profiles below ground level (mm)	1198
Height of back column profiles above ground level (mm)	2047
Height of back column profiles below ground level (mm)	1203



44 PVSPs were installed on the total covered area, A_{PVSP} which supported on 10 columns. High strength low carbon steels corresponding to the [Turkish European Standards TS EN 10279 \(2005\)](#) was noted as material quality in the design of PVSP steel frame. C-channel size of 125x62.5x25x4mm profiles made of galvanized cold-rolling S235JR (ASTM A283C) and S355JR (ASTM A441) steel material for the column and beam were considered, respectively.

Table 2. Material and geometrical properties of profiles used in PVSP steel frame

Designation of profiles	Profile name	Profile type	Material grade	<i>h</i> (mm)	<i>b</i> (mm)	<i>t</i> (mm)	Weight (N)
P1	Rail	Private design	S235JR	61	66	2.50	2765
P2	Beam	C-channel 125x62.5x25x4	S355JR	125	62.5	4	8584
P3	Front Column		S235JR				452
P4	Back Column		S235JR				880
P5	Purlin	C-channel 100x50x20x4	S235JR	100	50	4	150
P6	Brace	C-channel 50x25x5	S235JR	50	25	5	193

4. Codes and Standards

The standards used in the PVSPs steel structure project are the specification for buildings to be built in seismic zones ([Turkey Earthquake Codes \(TEC\), 2007](#)) (here named as Earthquake Regulation-2007) to find earthquake strength, [TS 498 \(1997\)](#) for the calculation values of wind and snow loads. 2019 Building Earthquake Code of Turkey has improved the previous version published in 2007 by expanding its coverage. Note that Earthquake Regulation-2007 was used here, as this study preceded the publication of the latest earthquake document of 2019 Building Earthquake Code of Turkey. Turkish steel design code [TS 648 \(1982\)](#) was used for the calculation and construction rules of steel structure. The principles of design of buildings, material intensity taken from [TS ISO 9194 \(1997\)](#). According to the used standards for design of PVSP support structure, combined with the basic parameters of the project, the typical dead load (*D*), wind load (*W*), snow load (*S*), and seismic load (*E*) of PVSP support structure were determined and presented in the subsequent sections.

5. Determining Design Loads and Load Combinations

The load of PVSP consisted of two parts such as (i) the dead load (*D*) included the weight of total PVSP (*D*₁), rail weight (*D*₂), beam (*D*₃), purlin (*D*₄), bolts (*D*₅), columns above ground level (*D*₆) (ii) the live load included wind load (*W*), snow load (*S*), and seismic load (*E*). The design loads and load combinations were explained in this section.

5.1. Determination of Wind Loads (*W*)

The procedures for TS 498 (TS 498-Chapter 11 – Wind Load Calculations) wind load procedure is briefly described below. Quantity of equivalent wind load is an area load which changes according to height of the structure. The design wind pressure loads *q_b* amounts are defined as 0.8*q_b* for pushing and 0.4*q_b* for suction. The wind creates 1.2*q_b* load on the top of structure in opposite direction of gravity if the angle of the roof with the horizontal plane is zero. Effective wind velocity, *v_b*, proposed by TS 498, is obtained as 38.9m/s. The PVSP is placed at an average height of 1.5m from the ground level. The air density, *ρ* is determined as 1.25kg/m³. As per TS 498 in Section 11.3, the design wind pressure can be obtained by using Eq. (1). The wind pressure coefficients incidence on top PVSP frame with the mounting angle, *α* = 25° are calculated and tabulated in Table 3.

$$q_b = \frac{1}{2} \rho v_b^2 \quad (1)$$

Table 3. Design wind pressure coefficients and wind loads on PVSP frame

Wind pushing/suction effect of PVSP	q_b (N/m ²)
$(1.2\sin\alpha-0.4)q_b$	101
$-0.4q_b$	-378
$0.8q_b$	756

5.2. Determination of Snow Loads (S)

Snowfall is unequally distributed in Turkey. Some regions should be adaptable to an additional challenge, namely the large masses of snowfall. The snow loading depends on the snow load zone, where the PVSP system will be installed. Due to the settling of snow on the panels, the solar modules and the mounting system can be highly loaded. The standard [TS 498 \(1999\)](#) (Chapter 7 – Snow Load Calculations) and [TS EN 1991-1-3 \(2007\)](#) include the methods of calculation for the snow load. Also, the calculation formulas for the snow load on pitched roofs are found in these standards. According to TS 498, the design ground snow load, P_{k0} , depends on the geographical location and the altitude of the site being considered. Due to the site of PV power plant is located on the land in Siirt province of south-eastern Turkey, design ground snow load at this site is taken as Zone II as per TS 498. The height of power plant site is accepted as 1325m above sea level. Additionally, in TS 498, the design roof snow load is calculated as the product of the design ground snow load and a conversion factor that depends on the roof slope. The snow loads applied vertically on the top of PVSP and the design ground snow load, P_{k0} is obtained as 880N/m². This load was uniformly distributed on the whole roof structure.

5.3. Determination of Seismic Loads (E)

Earthquake Regulation-2019 has improved the previous version published in 2007 by expanding its coverage. Note that Earthquake Regulation-2007 was used here, as this study preceded the publication of the latest earthquake document of Earthquake Regulation-2019. There are no differences between these two versions of Earthquake Regulation standards for earthquake zonation map of Siirt province. The earthquake hazard of Siirt province is considered in the first seismic zone. Determination of the local site soil conditions of PVSP support to be constructed based on the soil investigations was determined primarily in accordance with the rules and requirements of Earthquake regulation-2007. Z1 local site class for Siirt province was defined in accordance with the definition in Earthquake Regulation-2007. The building natural vibration period, T was obtained for the first mode shape as 0.178055sec from SAP2000. The spectral acceleration coefficient, $A(T)$, which shall be considered as the basis for the determination of seismic loads is given by Eq. (2). The spectrum coefficient, $S(T)$, seeming in Eq. (2) can be determined by Eq. (3), based on the local site conditions.

$$A(T) = A_0 \cdot I \cdot S(T) \quad (2)$$

$$S(T) = 1 + 1.5 \frac{T}{T_A} \quad (0 \leq T \leq T_A)$$

$$S(T) = 2.5 \quad (T_A \leq T \leq T_B) \quad (3)$$

$$S(T) = 2.5 \left(\frac{T_B}{T} \right)^{0.8} \quad (T_B < T)$$

Here, since the spectrum characteristic periods, T_A and T_B are specified depending on Z1 local site classes as $T_A = 0.10\text{sec}$ and $T_B = 0.3\text{sec}$, $S(T)$ is determined as 2.5 using by Eq. (3). The effective ground acceleration coefficient, A_0 , visible in Eq. (2) is defined as 0.40 for the first seismic zone. The building importance factor, I is specified as 1 assuming PVSP support structure like as industrial structure. By using these parameters, $A(T)$ is calculated as 1. The elastic seismic loads to be determined in terms of spectral acceleration coefficient defined in

Eq. (2) can be divided to below defined seismic load reduction factor to account for. Seismic load reduction factor, R_a is determined by Eq. (4) in terms of structural system behavior factor, R for various structural systems, and T . Since T in the project is less than T_A , $R_a(T) = R$ is taken as 4.

$$R_a(T) = 1.5 + (R - 1.5) \frac{T}{T_A} S \quad (0 \leq T \leq T_A)$$

$$R_a(T) = R \quad (T_A < T)$$
(4)

5.4. Design load combinations

In structural engineering practice, individual structural members are designed for the critical scenarios. Conventionally such critical scenarios are identified using structural analysis for different load combinations. Conventionally dead, live, earthquake and wind loads are the primary load types used to analyze a structure for various parameters. Further, combining load combinations and load patterns requires the engineer to do multiple iterations of structural analyses in order to capture the critical scenario. In fact, for simplicity standard structural engineering codes of practice have suggested several critical load patterns. Load combinations specified by TS 648 and Earthquake Regulation-2007 in Section 2.7.5 are tabulated in Table 4.

6. Analysis Method

The method to be used here for the seismic FEA of PVSP steel support structure by using [SAP2000 v14 \(2009\)](#) is equivalent seismic load method as described in Earthquake Regulation-2007. The analysis method can use for the PVSP support structure due to the height of frame is less than 40m. The distributed equivalent load is applied at the mass center of PVSP support structure. Firstly, for the specification of total equivalent earthquake loads, the total structure weights, W_T is calculated. Secondly, total equivalent seismic load (base shear), V_t , effective on the whole structure is specified by Eq. (5). This load is assigned to the mass center. The value of acceleration due to gravity, denoted g , is $g = 9.81m/s^2$. Here, total structure weight, W_T given in Eq. (5) is summarized in Table 5 and V_t is calculated using Eq. (5) as 21191.09N.

$$V_t = \frac{W_T A(T_1)}{R_a(T_1)} \geq 0.10 A_0 I W$$
(5)

Table 4. Classification of load combinations

Load condition		Load combinations
Long-term	The usual	D+0.7S
Short-term	Snow	D+S
	Wind	D+S+W
		D+S-W
		0.9D+W
		0.9D-W
	Earthquake	D+S+E
		D+S-E
		0.9D+E
		0.9D-E

Table 5. Total structure weight, W_T

Definition of weights	Designated load cases	W_T (N)
Total weight of PVSP	D_1	$18.5 \times 4 \times 11 \times 9.81 = 7985.34$
Total weight of top frame (rail+beam+purlin+bolt)	$D_2+D_3+D_4+D_5$	$2765+8584+150+425=11924$
Total weight per a column (front and back) above ground level	D_6	$(452+880)/4=333$
Snow weight on A_{PVSP}	S	$880 \times 73.32 = 64522$
Total structure weight		$\Sigma W_T = 84764.34$

The rules for dimensioning of the frames with normal ductility level are given in Earthquake Regulation-2007. Each load bearing members' section should be checked for local buckling which is defined as a slenderness ratio of the width to flange thickness in C-channel sections by considering the ductility level in the design. In this paper, the requirement for a compact C-channel sections were fulfilled by all frame' sections for the normal ductility level, because the upper limit values given in Eq. (6) and Eq. (7) as below were satisfied, where b and t are the width and thickness of C-channel section (see in Table 2), and E_s is the elasticity modulus of the steel material considered as $210GPa$.

$$\frac{b}{t} \leq 0.4 \sqrt{(E_s / \sigma_a)} \quad (6)$$

$$\frac{h}{t} \leq 4.0 \sqrt{(E_s / \sigma_a)} \quad (7)$$

7. Evaluation of Results

The model was designed for the critical scenarios to obtain the resulting von Mises stress distributions on the structure which obtained from SAP2000. It was observed that the maximum von Mises stresses σ_{max} was obtained as $210.5N/mm^2 < 212N/mm^2$ for steel material of S355JR and $114N/mm^2 < 144N/mm^2$ for steel material of S235JR which fulfilled the strength requirements for beam and column with the appropriate structure safety. According to the force analysis results of the model in SAP2000, the maximum axial force obtained among the columns of 10 was $11670.44N$ as seen in Figure 3. This value was used as the maximum shear force to check the strength of bolts. As mentioned before, 2M18 steel bolts used in each connection between column and beam were designed per the [TS EN 1993-1-8 \(1993\)](#). The strength of frame and bolts were controlled, and it fulfilled the requirements for the PVSP steel support structure.

8. Conclusions

The purpose of this paper is to discuss the structural design of PV systems for wind, snow, earthquake loads in the Turkey, and provide guidance using current Turkish codes as appropriate. Codes and standards thereon provide a key source representing the state-of-the-art information on a given aspect of engineering discipline enabling the extension of knowhow, based on research and experience, into practice. In view of the increasingly improved database on PVSPs is great help to a designer. This paper has been developed for this purpose. With the introduction of the PV solar power plant, potential design principles used for calculating wind, snow and earthquake loads for PV systems in the Turkey is provided.

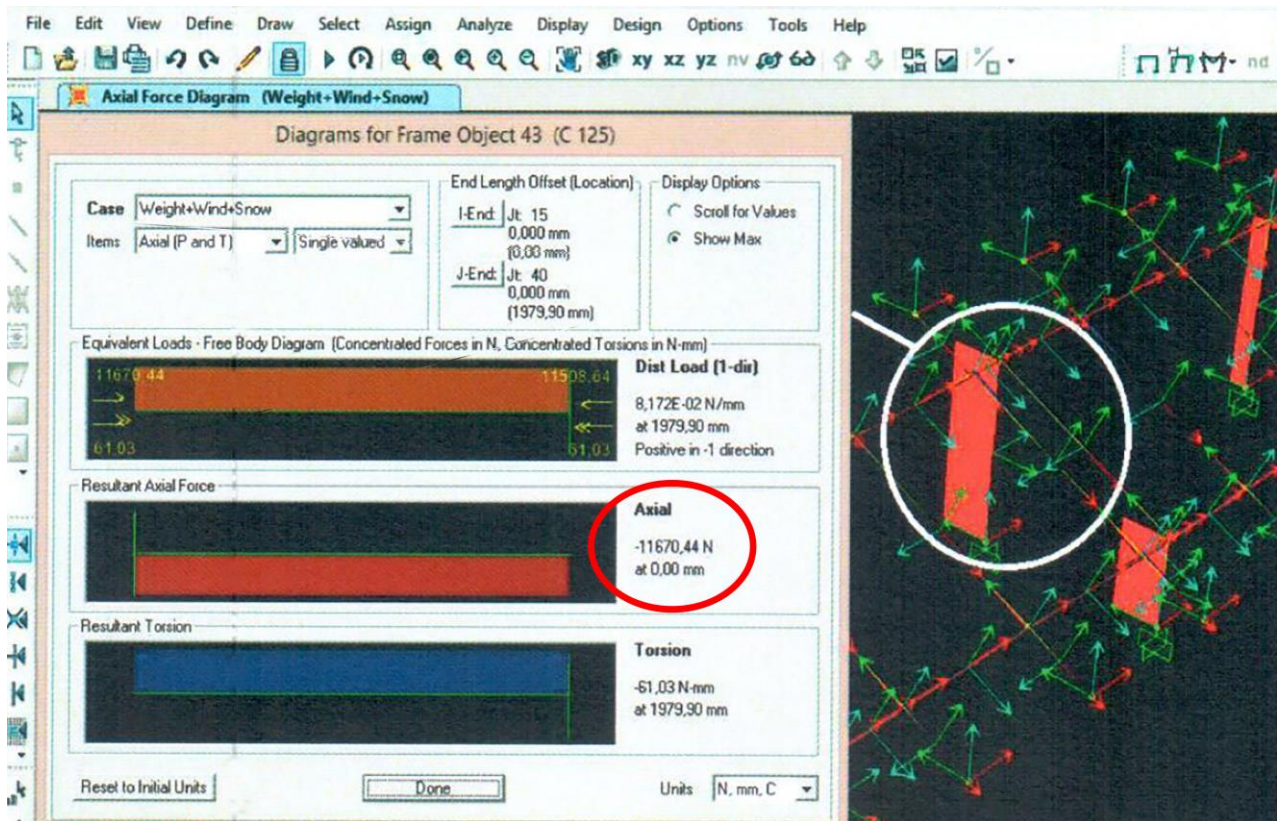


Figure 3. The maximum axial force to check bolts

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